

Node.js / Full Stack Interview Quick Revision

Questions & Answers discussed today

1. Authentication vs Authorization

Authentication verifies who the user is. Authorization verifies what the user can access. JWT is commonly used for authentication and RBAC for authorization.

2. Redis vs Database

Database is the source of truth. Redis is used as a cache for frequently accessed data to improve performance.

3. Indexing

Indexing speeds up data retrieval. Common indexes: Single Field, Compound, Unique, Text, TTL. Search complexity is typically $O(\log n)$.

4. Sharding

Sharding horizontally partitions data across multiple servers. Common strategies: Range, Hash, Directory, Geographic.

5. Sharding Problems

Cross-shard joins, hotspotting, rebalancing complexity, distributed transactions, operational overhead.

6. Event Driven Architecture

Services communicate via events using Kafka, RabbitMQ, or SQS. Benefits include loose coupling and scalability.

7. AWS SQS

Managed message queue service used for asynchronous processing. Common use cases: emails, notifications, background jobs.

8. SQS Architecture

Producer -> SQS Queue -> Consumer/Lambda -> Database or External Service.

9. DLQ (Dead Letter Queue)

Failed messages are moved to DLQ after exceeding retry attempts, helping debugging and recovery.

10. Kafka vs SQS

Use SQS for simple AWS-based queues and background jobs. Use Kafka for high-throughput event streaming and multiple consumers.

11. gRPC

High-performance communication protocol using HTTP/2 and Protocol Buffers. Commonly used for microservice communication.

12. URL Shortener Design

Generate unique short code, store mapping in DB, cache frequently accessed URLs in Redis, and redirect using 302.

13. DSA Complexity Quick Notes

HashMap Lookup: $O(1)$, B-Tree Index Search: $O(\log n)$, Array Search: $O(n)$, Binary Search: $O(\log n)$.